DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING						
Course Code: 24CS320		Course Title: *Digital System Design*				
Total Credits: 04 *L-45: T-0: P-30: SL and TW-45*		*SDGs addressed: 4, 8, 9*	*Course Category: PCC*		*Course Type: Theory with Practical*	
Teaching Learning Process		*Total Contact Hours/ Semester*	*Assessment in Weightage and marks*			
	Lecture	45		CIE	SEE	Total
	Tutorial	-	*Weightage in %*	*40*	*60*	*100*
	Practical	30	*Maximum Marks*	*40*	*60*	*100*
	Self-learning and Term Work	45	*Minimum Marks*	*20*	*25*	*45**
	Total	120				

Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)

Course Prerequisite: NIL

Course Objectives:

Sl. No.	Course Objectives
1	Understand Boolean algebra and basic properties of Boolean algebra; able to simplify simple Boolean functions by using the basic Boolean properties.
2	Able to design simple combinational logics using basic gates. Able to optimize simple logic using Karnaugh maps, understand "don't care".
3	Familiar with basic sequential logic components: SR Latch, D Flip-Flop and their usage and able to analyze sequential logic circuits.
4	Understand finite state machines (FSM) concepts.

Course Outcomes (COs)

CO	Course Outcomes	Highest Level of Cognitive Domain	Working Knowledge
CO1	Explain the fundamental concepts of Boolean algebra, Boolean theorems, K-MAP, Quine-McCluskey algorithm and apply the knowledge of theoretical foundations to simplify and solve Boolean functions.	L3	WK1, WK2
CO2	Design and implement the combinational circuits.	L3	WK1, WK2
CO3	Apply the knowledge of timing aspects and operation of Flip-Flops in conversion between different Flip Flops.	L3	WK1, WK3
CO4	Illustrate the working of various registers and register circuits.	L2	WK1, WK3
CO5	Design and implement synchronous/asynchronous counters and synchronous sequential circuits.	L3	WK1, WK6

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

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Course Content / Syllabus:

UNIT No.	Content	Lecture Hours
1	**Digital Logic:** Combinational Logic Circuits, Boolean laws and Theorems, Sum-of-Products Method, Truth Table to Karnaugh Map, Pairs, Quads, and Octets, Karnaugh Simplifications, Don't-care Conditions, Product-of-sums Method, Product-of-sums simplifications, Simplification by Quine- McCuskey Method. **Self-Learning and Term Work*** • Number System • Axiomatic Definition of Boolean Algebra or Huntington Postulates	09
2	**Data Processing Circuits:** Multiplexers, Demultiplexers, 1-of-16 Decoder, BCD to Decimal Decoders, Encoders, Parity Generators and Checkers, Magnitude Comparator. **Self-Learning and Term Work*** • Parallel adder with look-ahead method • Conversion from one combinational circuit to another.	09
3	**Flip-Flops**: RS Flip-Flops, Gated Flip-Flops, Edge-triggered RS FLIP-FLOP, Edge-triggered D FLIP-FLOPs, Edge-triggered JK FLIP-FLOPs, JK Master-slave FLIP-FLOP, Switch Contact Bounce Circuits, Various Representation of FLIP-FLOPs, Conversion of Flip-Flops. **Self-Learning and Term Work*** • Designing of different flip-flops using state tables and state diagrams. • Flip-flops and Counter design.	09
4	**Registers:** Types of Registers, Serial In - Serial Out, Serial In - Parallel out, Parallel In - Serial Out, Parallel In - Parallel Out, Universal Shift Register, Applications of Shift Registers. **Self-Learning and Term Work*** • Design process of registers, clock synchronization, bit shifting, and resetting mechanisms	09
5	**Counters:** Asynchronous Counters, Synchronous Counters, Decade Counters. **Design of Synchronous Sequential Circuits**: Model Selection, State Transition Diagram, State Synthesis table, Design equations and Circuit diagram. **Self-Learning and Term Work*** • Design of Mod-N counters. • Study of Basic Moore and Mealy Machine.	09

*** NOTE: Self-Learning and Term Work are evaluated only through continuous internal evaluation (CIE) and will not be assessed during the Semester End Examination (SEE).**

Text Book:

Sl. No.	Author/s	Title	Publisher Details
1	Donald P Leach, Albert Paul Malvino and Goutam Saha	Digital Principles and Applications	7th Edition, Tata McGraw Hill, 2014.

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Reference Books:

Sl. No.	Author/s	Title	Publisher Details
1	Morris Mano, Kim and Charles	Logic and Computer Design Fundamentals	5th Edition, Prentice Hall, 2016.
2	Donald D.Givone	Digital Principles and Design	1st Edition, Tata McGraw-Hill, 2012.
3	John M.Yarbrough	Digital Logic Applications and Design	1st Edition, Thomson Learning, 2001
4	Kohavi	Switching and Finite Automata Theory	3rd Edition, TMH, 2010.

Web Resources:

Sl. No.	Web Links
1	https://nptel.ac.in/courses/108105113/15
2	https://nptel.ac.in/courses/117105080/19

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	3	-	-	-	-	-	-	-	-	3	-	-
CO2	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO4	3	3	-	-	-	-	-	-	-	-	-	3	3	3
CO5	3	3	3	-	-	-	-	-	-	-	-	3	3	-

High – 3, Medium – 2, Low – 1

LIST OF PROGRAMS FOR PRACTICAL SESSIONS

Weeks	List of Programs	No. of Hours
1	a. Study and verify the truth table of logic gates. b. Realize Basic Gates using Universal Gates. c. Simplify the given expression and realize it using gates.	02
2	Design and realize a. Half Adder / Full Adder. b. Half Subtractor / Full Subtractor.	02
3	Design and set up the following circuit using IC 7483. a. A 4-bit binary parallel adder. b. A 4-bit binary parallel subtractor.	02
4	Design and realize the following using Basic gates & IC 7483. a. BCD to Excess- 3Code and vice versa b. Binary to Gray Code and vice versa	02
5	Design and implement a. 4:1 Multiplexer (MUX) using only NAND gates. b. Boolean expressions using 8:1 Multiplexer(MUX).	02
6	Half/Full Adder and Half/Full Subtractor using IC 74153.	02
7	Design and implement One & Two-Bit Magnitude Comparator.	02
8	Design and implement a. Decoder circuit using basic gates and to verify using IC74LS138.	02

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	b. Half/Full Adder and Half/Full Subtractor using IC74LS138.	
9	a. Design and implement 3-bit parity generator and verify using Parity Checker. b. Implementation and verification of truth table for J-K flip-flop, Master-slave J- K flip-flop, D flip-flop and T flip-flop.	02
10	Design and implementation of Mod-N synchronous counter using J-K flip-flops.	02
11	Design and implementation of shift register to function as i) SISO ii) SIPO iii) PISO iv) PIPO v) shift left and vi) shift right operation.	02
12	a. Design and implementation of i) Ring counter and ii) Johnson counter using 4- bit shift register. b. Design and implementation of Decade Counters c. Design and implementation of Asynchronous Counters	02
13	**Revision**	02
14	**Revision**	02
15	**Laboratory Test/Event**	02

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